

# FPGA-Based Real-Time Driver Drowsiness Detection Using RISC-V and Hardware Acceleration

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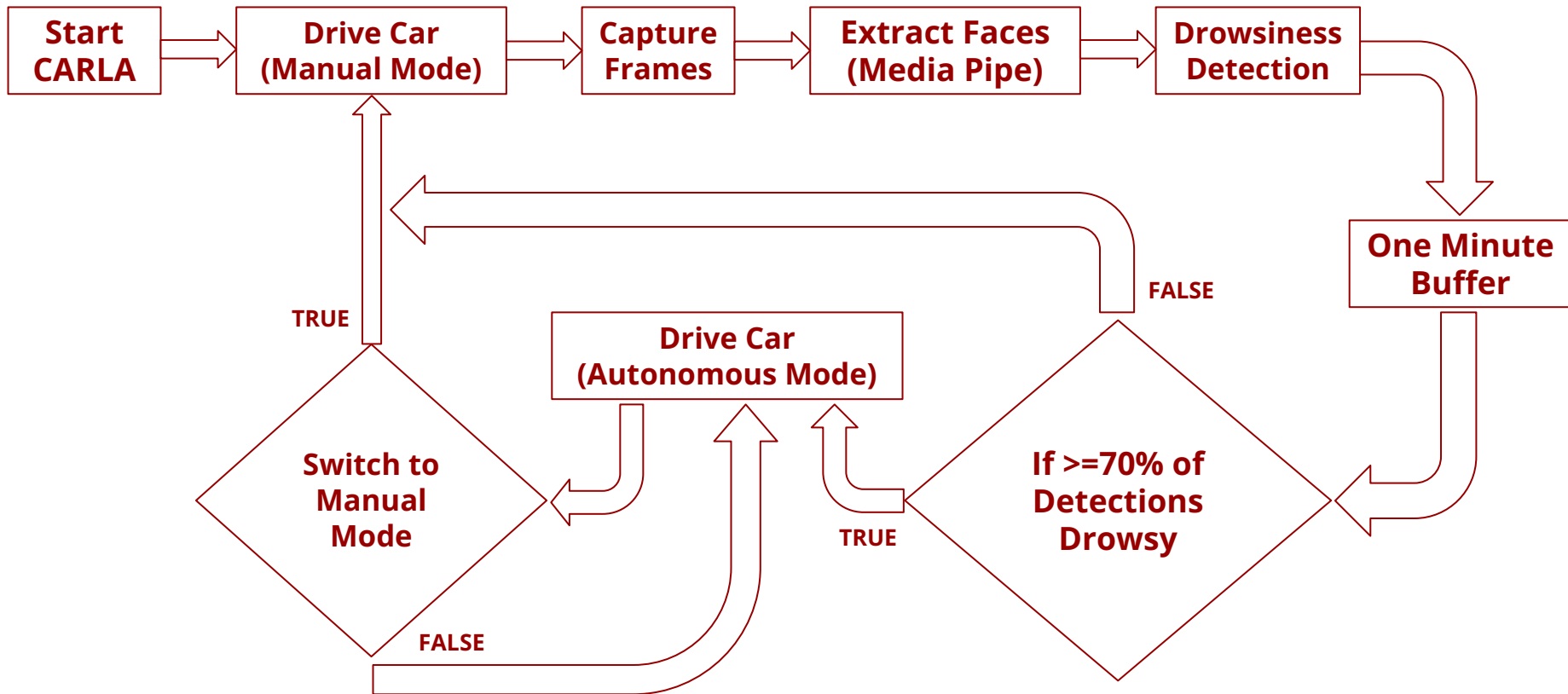
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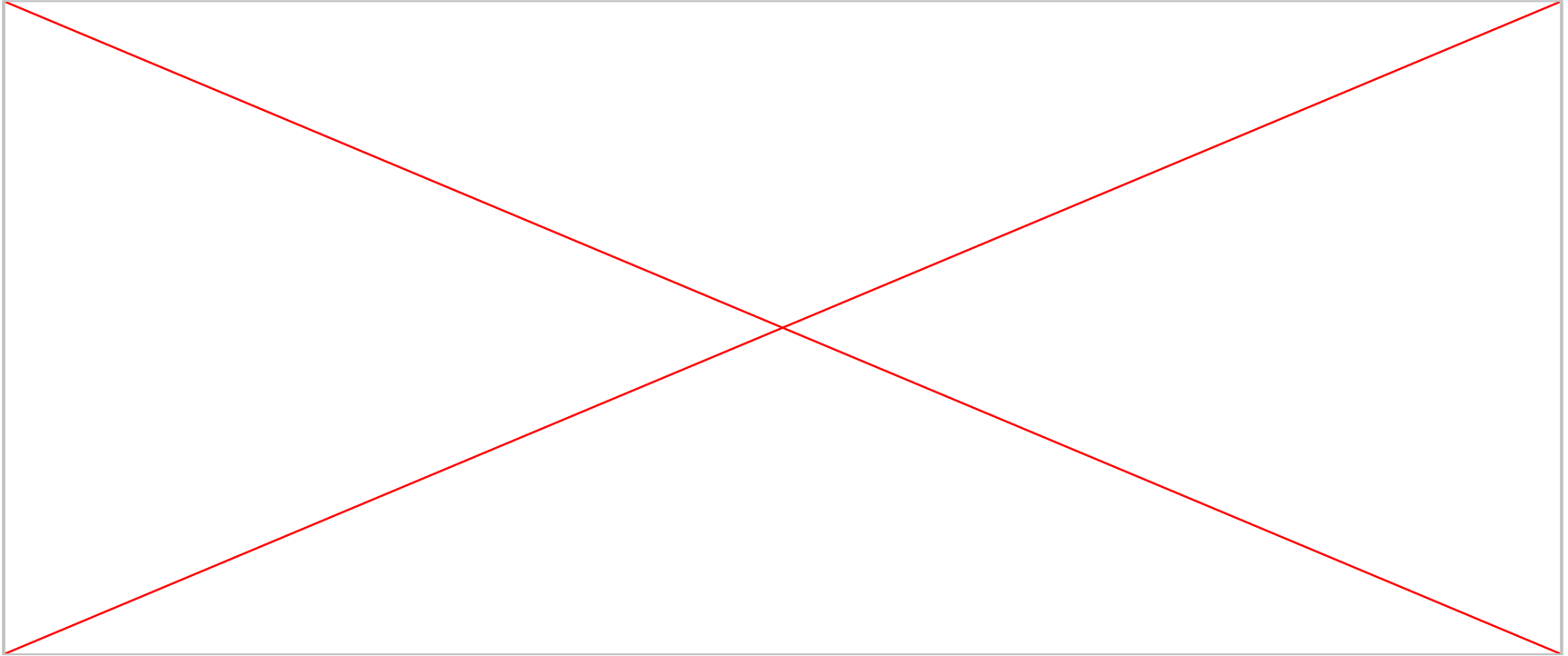
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# System Architecture (Software)



# Software Demo



# Optimization Pipelines

## Standard Optimization Pipeline (TFMOT)

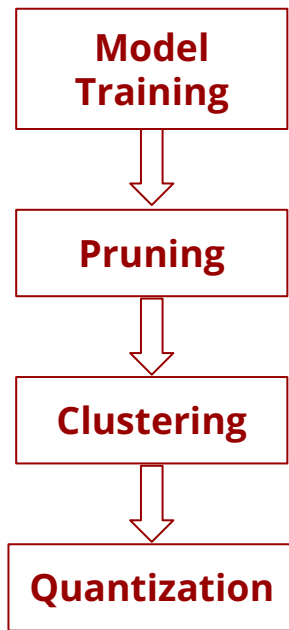
### Model Size Reduction

**Model Compression** : Achieved **6x size reduction** ↑ via the TFMOT pipeline

**The Bottleneck** : Observed **17x higher latency** ↓ and **11x lower throughput** ↓ in Keras

**The Solution** : Migrated to **ONNX** to **bypass high-level software overhead**

**Hardware Fusion** : **Combined model layers** to restore real-time performance for FPGA



## ONNX Optimization Pipeline

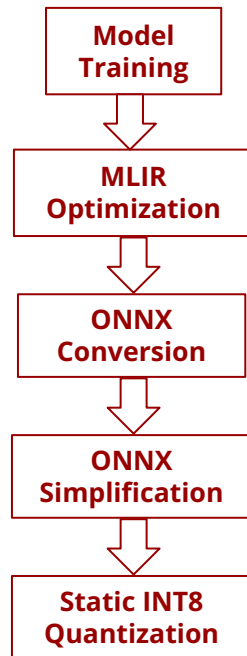
### Latency Reduction

**Performance Gain**: Achieved **8x higher throughput** ↑ and **6.5x lower latency** ↑

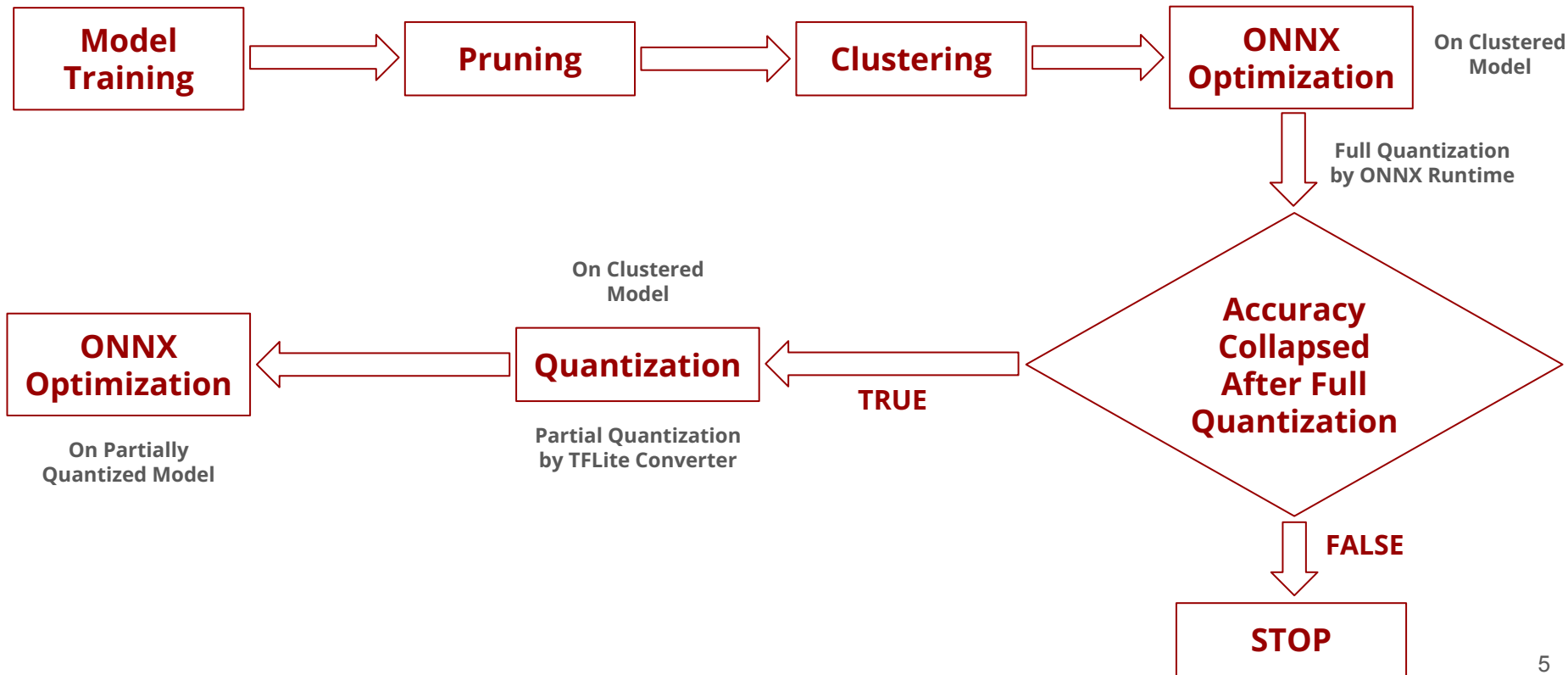
**Accuracy Collapse**: After Quantization, Observed a drop from **99% to 55%** ↓ with marginal throughput and latency degradation

**The Solution**: Hybridized **TFLite Quantization** with the **ONNX pipeline**

**Expected Result**: Restored precision with **90%+ accuracy** retention for FPGA



# Merged Optimization Pipeline



# Performance Report (Partially Quantized Models)

| <i>MobileNetV3 Small</i> | Throughput                | Accuracy      | Model Size (Raw) | Model Size (ZIP) | Latency         |
|--------------------------|---------------------------|---------------|------------------|------------------|-----------------|
| Original                 | 24.1<br>Imgs/sec          | 99.14%        | 31.20 MB         | 28.24 MB         | 41.54 ms        |
| Pruning                  | 16.7<br>Imgs/sec          | 99.14%        | 12.50 MB         | 7.09 MB          | 59.73 ms        |
| Clustering               | 15.6<br>Imgs/sec          | 98.90%        | 12.38 MB         | 1.90 MB          | 64.17 ms        |
| Partial Quantization     | 9.73<br>Imgs/sec          | 93.83%        | 3.22 MB          | 1.53 MB          | 102.73 ms       |
| Simplified FP32 ONNX     | <b>80.38<br/>Imgs/sec</b> | <b>96.88%</b> | <b>3.18 MB</b>   | <b>1.48 MB</b>   | <b>12.44 ms</b> |
| Fully Quantized ONNX     | 202.52<br>Imgs/sec        | <b>47.16%</b> | 3.21 MB          | 1.49 MB          | 4.94 ms         |

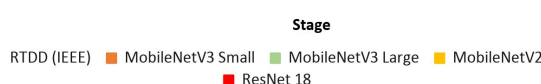
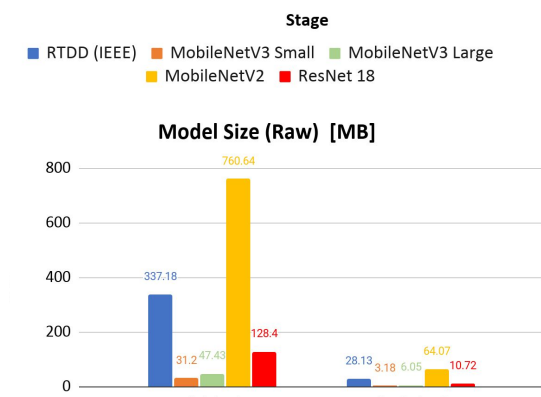
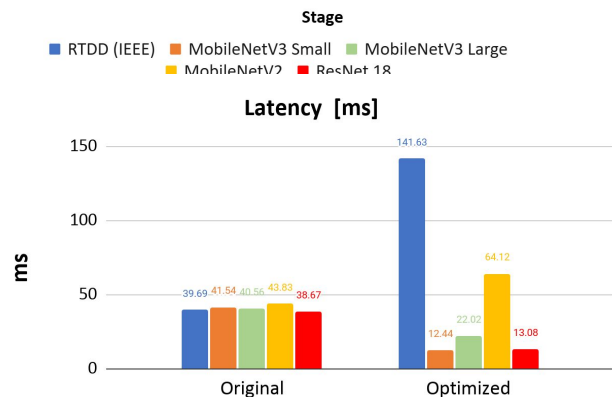
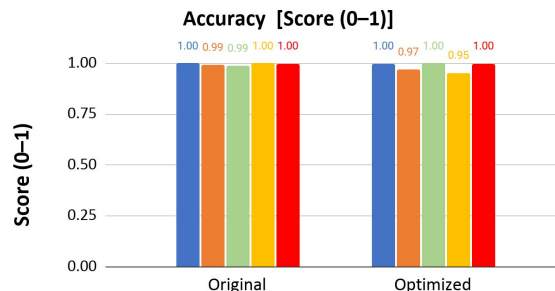
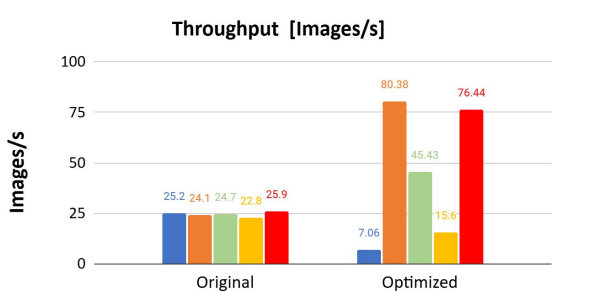
| <i>MobileNetV3 Large</i> | Throughput                | Accuracy       | Model Size (Raw) | Model Size (ZIP) | Latency         |
|--------------------------|---------------------------|----------------|------------------|------------------|-----------------|
| Original                 | 24.7<br>Imgs/sec          | 98.76%         | 47.43 MB         | 41.17 MB         | 40.56 ms        |
| Pruning                  | 14.9<br>Imgs/sec          | 98.76%         | 23.96 MB         | 16.95 MB         | 67.30 ms        |
| Clustering               | 14.8<br>Imgs/sec          | 99.90%         | 23.81 MB         | 4.24 MB          | 67.36 ms        |
| Partial Quantization     | 3.17<br>Imgs/sec          | 76.37%         | 6.22 MB          | 3.63 MB          | 315.50 ms       |
| Simplified FP32 ONNX     | <b>45.43<br/>Imgs/sec</b> | <b>100.00%</b> | <b>6.05 MB</b>   | <b>3.50 MB</b>   | <b>22.02 ms</b> |
| Fully Quantized ONNX     | 119.61<br>Imgs/sec        | <b>52.92%</b>  | 6.07 MB          | 3.26 MB          | 8.36 ms         |

# Performance Report (Fully Quantized Models)

| <i>ResNet - 18</i>          | Throughput                | Accuracy      | Model Size (Raw) | Model Size (ZIP) | Latency         |
|-----------------------------|---------------------------|---------------|------------------|------------------|-----------------|
| <b>Original</b>             | 25.9<br>Imgs/sec          | 99.86%        | 128.40 MB        | 118.27 MB        | 38.67 ms        |
| <b>Pruning</b>              | 25.7<br>Imgs/sec          | 99.86%        | 42.89 MB         | 39.61 MB         | 36.37 ms        |
| <b>Clustering</b>           | 45.8<br>Imgs/sec          | 99.90%        | 42.87 MB         | 9.39 MB          | 21.84 ms        |
| <b>Simplified FP32 ONNX</b> | 56.44<br>Imgs/sec         | 99.58%        | 42.64 MB         | 39.53 MB         | 17.72 ms        |
| <b>Fully Quantized ONNX</b> | <b>76.44<br/>Imgs/sec</b> | <b>99.79%</b> | <b>10.72 MB</b>  | <b>10.15 MB</b>  | <b>13.08 ms</b> |

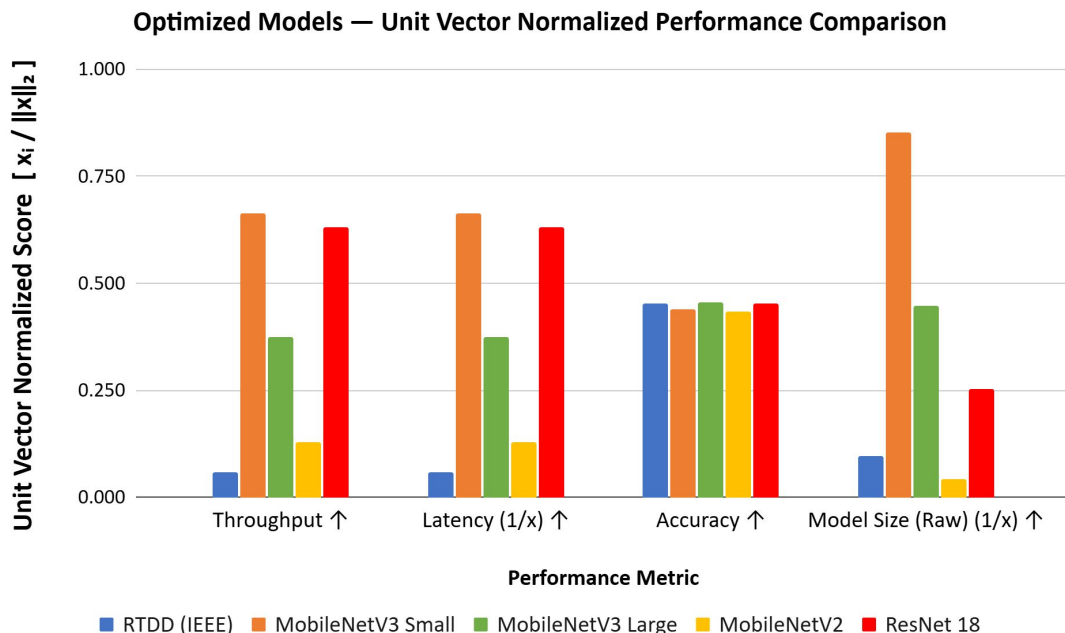
| <i>RTDD (IEEE)</i>          | Throughput               | Accuracy      | Model Size (Raw) | Model Size (ZIP) | Latency          |
|-----------------------------|--------------------------|---------------|------------------|------------------|------------------|
| <b>Original</b>             | 25.2<br>Imgs/sec         | 99.90%        | 337.18 MB        | 312.66 MB        | 39.69 ms         |
| <b>Pruning</b>              | 135.3<br>Imgs/sec        | 99.81%        | 112.42 MB        | 39.20 MB         | 7.39 ms          |
| <b>Clustering</b>           | 135.1<br>Imgs/sec        | 99.76%        | 112.41 MB        | 7.94 MB          | 7.40 ms          |
| <b>Simplified FP32 ONNX</b> | 6.08<br>Imgs/sec         | 99.79%        | 112.37 MB        | 39.19 MB         | 164.57 ms        |
| <b>Fully Quantized ONNX</b> | <b>7.06<br/>Imgs/sec</b> | <b>99.79%</b> | <b>28.13 MB</b>  | <b>10.76 MB</b>  | <b>141.63 ms</b> |

# Performance Gains Across Models



- Throughput** : Three of five models achieved substantial throughput gains following optimization. **MobileNetV3 Small** delivered the highest improvement at **+233%** (24.1 → 80.38 img/s), followed by **ResNet 18** at **+195%** (25.9 → 76.44 img/s) and **MobileNetV3 Large** at **+84%** (24.7 → 45.43 img/s).
- Latency** : Inference latency was significantly reduced in three models. **MobileNetV3 Small** achieved a **70% latency reduction** (41.54 → 12.44 ms), **ResNet 18** a **66% reduction** (38.67 → 13.08 ms), and **MobileNetV3 Large** a **46% reduction** (40.56 → 22.02 ms), making them viable for real-time edge deployment.
- Model Size Reduction** : All five models achieved drastic reductions in model size, averaging **over 90% compression** of raw model size. The most significant reduction was seen in **MobileNetV2**, shrinking from **760.64 MB to 64.07 MB (–92%)**, followed by **RTDD (IEEE)** from **337.18 MB to 28.13 MB (–92%)** and **ResNet 18** from **128.40 MB to 10.72 MB (–92%)**.

# Optimized Model Selection for Edge Deployment



- Multi-Metric Benchmarking** : All five optimized models were compared across Throughput, Latency, Accuracy, and Model Size using unit vector normalization for fair cross-metric evaluation. **MobileNetV3 Small**, **MobileNetV3 Large**, and **ResNet 18** consistently ranked as the top three performers.
- MobileNetV3 Small — Best Overall Performer** : **MobileNetV3 Small** ranked best in 3 out of 4 metrics, achieving the **highest throughput (80.38 img/s)**, **lowest latency (12.44 ms)**, and **smallest model size (3.18 MB)**, making it the most suitable model for edge deployment.
- ResNet 18 — Final Selection for Hardware Acceleration** : **ResNet 18** was ultimately selected for deployment on our custom accelerator built using **Apache TVM**. While **MobileNetV3 Small** operates in **mixed precision** — making TVM acceleration code **significantly more complex** to write — **ResNet 18's fully quantized architecture greatly simplified the implementation of the acceleration pipeline**, making it the more practical choice for our custom hardware.

# TVM Migration to Shakti



## Why TVM?

- **Initial Setup (TFLM on Shakti CPU) :**

The TensorFlow Lite Micro (TFLM) framework was initially ported to the Shakti CPU platform.

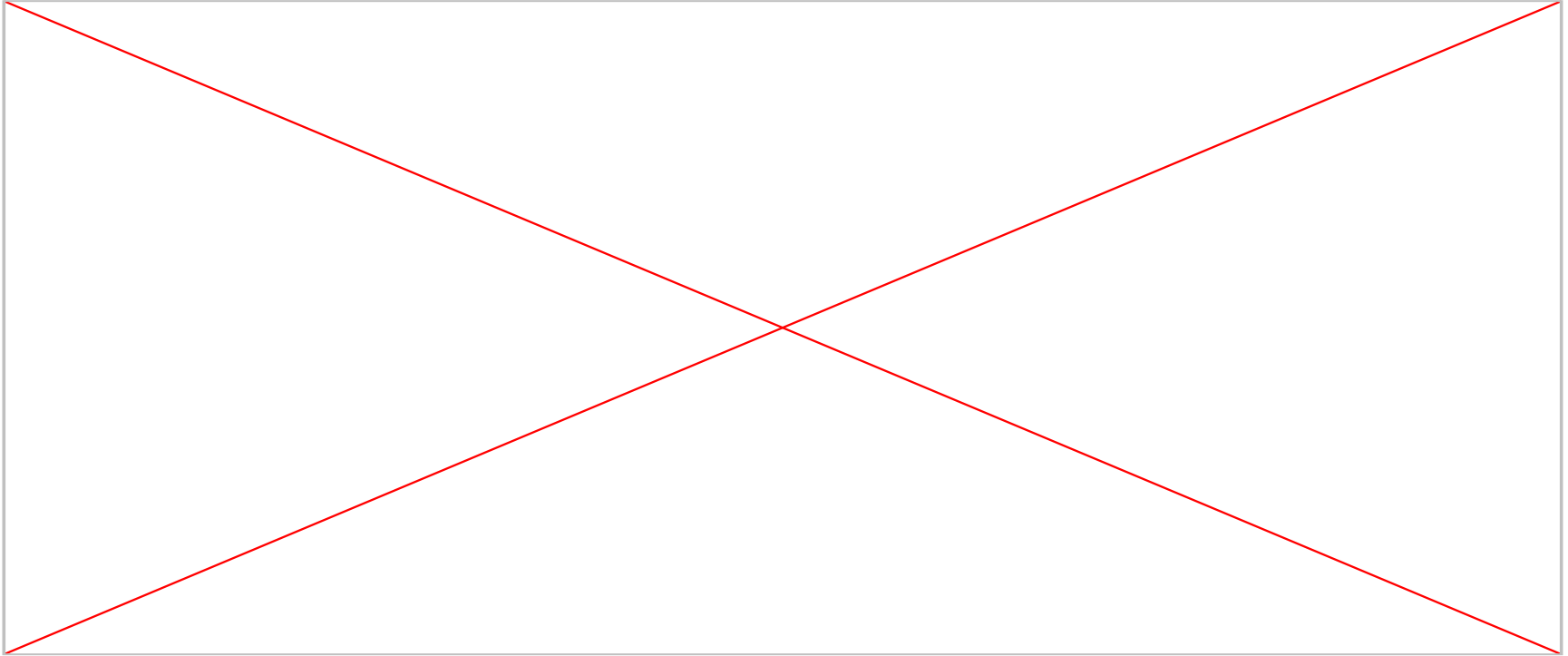
- **INT8 Quantization Limitation (TFLM) :**

Applying INT8 quantization led to a significant accuracy drop (~99% → ~52%) so switched to mixed precision support.

- **Approach :**

- **Switched to TVM :** Since mixed precision models are not supported by TFLM framework.
- **Graph Optimization :** Applied transformations such as *SimplifyInference* and *FoldConstant* for improved efficiency.

# TFLM Demo



# TVM Demo

